

Hawkeye: Vision-Based UAV-Gimbal Target Tracking with Multi-Objective NPSO-Tuned State-Space PD Control

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Abstract

Real-time target tracking and active visual servoing on mobile robotic platforms demand a tightly integrated pipeline that balances detection precision, state estimation, and dynamic controller response. In this work, we propose a closed-loop visual control architecture that seamlessly couples real-time object detection and tracking with a gain-optimized proportional-derivative control mechanism (SS-PD). Operating within a ROS 2 framework and evaluated in NVIDIA Isaac Sim, the perception pipeline extracts high-frequency bounding box coordinates to formulate normalized visual error metrics, which are directly fed into the dynamic controller. To address systemic overshoots and external disturbance variations during rapid target maneuvers, we optimize the control parameters using a particle-swarm based gain-tuning strategy. Experimental evaluations demonstrate reduced transient overshoot and improved line-of-sight regulation relative to the tested PD baseline.

Keywords: UAV gimbal tracking; visual servoing; target tracking; state-space PD control; particle swarm optimization; NPSO tuning; YOLOv8–ByteTrack; ROS 2; NVIDIA Isaac Sim

1. Introduction

Unmanned aerial vehicles (UAVs) equipped with camera gimbals are increasingly used for surveillance, search-and-rescue, infrastructure inspection, and environmental monitoring. In these applications, the camera must maintain a stable line of sight to a target while the platform and target move relative to one another. This requirement is naturally formulated as an image-based visual servoing problem [1,2], in which image features are used directly in the feedback loop [15,17] to regulate the camera orientation and keep the target close to the image centre [16].

Recent UAV tracking systems combine object detection with active gimbals to maintain target visibility [5,19], while related work addresses autonomous moving-target tracking [22] and line-of-sight control under platform motion and disturbances [20,23]. Deep detectors provide image-plane target localization [4,24], whereas tracking-by-detection methods support temporal association [8,25,26]. Nevertheless, stable tracking also requires rejection of target- and carrier-induced disturbances under perception latency, detector noise, finite sampling, association errors, field-of-view limits, and actuator constraints.

These practical effects introduce competing control requirements. An aggressive controller may reduce the average image-plane tracking error, but can also increase transient overshoot and produce oscillatory gimbal responses [20,23]. Rapid or insufficiently filtered corrective motion can further degrade line-of-sight stability and appear as visible

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image jitter [21,35]. Conversely, a conservative controller may provide smoother camera motion while allowing larger target-centering errors. Controller tuning for vision-based gimbal tracking should therefore consider not only average tracking accuracy, but also the oscillatory behaviour, and transient characteristics of the closed-loop response.

Proportional–integral–derivative (PID) control remains attractive for gimbal stabilization and tracking because of its simple structure, low computational cost, and robustness [9,10]. Classical PID and PD controllers can provide acceptable gimbal-tracking performance [9,10], but their responses depend strongly on the selected controller gains [11,12]. The Ziegler–Nichols method provides a straightforward heuristic procedure for selecting these gains [27]. However, such heuristic tuning does not explicitly optimize competing requirements such as target-centering accuracy, smoothness, and transient overshoot [11,12]. Alternative gimbal-control strategies have also been investigated, including self-tuning fuzzy PID control [33], adaptive feedback linearization [34], and filter-assisted stabilization methods [35]. State-space PID (SS-PID) provides an observer-based realization that connects classical PID action with a compact state-space structure [6]. Its observer bandwidth provides an additional design parameter, making the formulation suitable for sampled vision-control systems. However, its performance still depends on the joint selection of the controller gains and observer parameters.

Metaheuristic optimization provides a practical alternative when the closed-loop objective is nonlinear, nonconvex, or difficult to express analytically. Particle swarm optimization (PSO) was originally introduced as a population-based, derivative-free search method [3]. Subsequent developments introduced inertia weighting to improve the balance between global exploration and local exploitation [28], while the stability and convergence properties of the particle dynamics were further addressed through the constriction factor [29].

PSO has also been extended to multi-objective problems, in which a set of Pareto-optimal solutions is sought for competing performance objectives [30]. For controller design, PSO has been applied to automatic PID parameter selection in general control systems [31,32]. In UAV-gimbal applications, conventional PSO has been used to tune a PID controller for a two-axis camera gimbal [9], while a new particle-swarm optimization PID (NPSO-PID) method has been developed for a two-axis stabilized gimbal [7].

However, many PSO-based PID tuning approaches rely on a single scalar error criterion, such as the integral of squared error, the integral of time-weighted absolute error, or the mean tracking error [11,12]. Although such criteria can improve average tracking accuracy, they do not directly account for oscillation, abrupt error variation, or transient overshoot. Comparatively less attention has been given to incorporating complementary image-domain quality measures into state-space PD tuning for UAV-gimbal tracking and to characterizing the resulting accuracy–smoothness trade-offs.

Simulation environments provide a safe and repeatable means of developing perception–control pipelines before hardware deployment. NVIDIA Isaac Sim supports photorealistic rendering, robot articulation, programmable scene logic, and ROS 2 integration, making it suitable for controlled closed-loop robotic-vision experiments [13]. ROS 2 provides a modular communication layer between perception, control, and actuation nodes [14]. Such an environment enables controller variants to be evaluated under identical carrier and target trajectories, thereby reducing uncontrolled experimental variation.

The main contributions of this work are as follows:

1. An end-to-end closed-loop simulation framework for vision-based UAV-gimbal target tracking that integrates YOLOv8–ByteTrack perception, ROS 2 communication, NVIDIA Isaac Sim actuation, image-based angular feedback, and SS-PD gimbal control.

2. A multi-objective NPSO formulation for joint selection of the SS-PD gains and observer bandwidth using fitness components based on mean absolute tracking error, error volatility, acceleration-like second-order error variation, and zero-crossing count.
3. A controlled comparison between classical Ziegler–Nichols PD and state-space PD realizations, followed by a systematic evaluation of NPSO-tuned controller variants under two complementary scenarios: continuous tracking with carrier- and target-induced line-of-sight disturbances and transient recovery from a large initial image-plane offset.
4. An empirical characterization of the accuracy–smoothness trade-off produced by the different fitness terms, identifying configurations that favour tight target centering, reduced image-centre crossings, lower transient overshoot, or smoother visual-error evolution.

The remainder of the paper is organized as follows. Section 2 formulates the visual tracking problem. Section 3 presents the perception–control methodology, including target observation, image-plane error extraction, pixel-to-angle conversion, and constrained gimbal actuation. Section 4 details the state-space PD realization and offline NPSO tuning procedure. Section 5 describes the experimental environment, evaluation scenarios, and assessment protocol. Section 6 reports the experimental results, while Section 7 discusses their implications, associated trade-offs, and limitations. Finally, Section 8 concludes the paper.

2. Problem Definition

The objective of vision-based UAV gimbal tracking is to continuously orient a camera-mounted gimbal such that a selected moving target remains close to the centre of the image plane [19,20]. We consider a monocular RGB camera rigidly mounted on a yaw–pitch gimbal carried by a UAV. Both the carrier and the target may move, thereby producing time-varying disturbances in the camera line of sight [20,21]. The target is regulated directly from image measurements without explicit depth estimation, three-dimensional reconstruction, or prior knowledge of its geometry. As illustrated in Figure 1, at each valid observation, the perception module must estimate the target location in the image plane. The difference between this location and the image centre is converted into angular line-of-sight feedback and supplied to the gimbal controller.

The control task is to generate physically admissible yaw and pitch commands that drive the image-plane tracking error towards zero despite carrier motion, target motion, perception uncertainty, and finite sampling [19,20]. In addition to accurate target centering, the closed-loop response should avoid excessive overshoot, abrupt corrections, and repeated oscillation around the optical axis, which are important considerations for stabilized optical imaging systems [21]. The corresponding controller parameters are therefore selected by balancing tracking accuracy, temporal smoothness, transient behaviour, and oscillation suppression over the admissible parameter space.

3. Perception–Control Methodology

The proposed methodology implements a closed-loop image-based visual servoing framework for human-target tracking using a camera mounted on a UAV gimbal. The framework combines NVIDIA Isaac Sim, ROS 2 communication, target detection and temporal association, image-plane error extraction, angular feedback generation, and constrained gimbal actuation. Its processing sequence is summarized as shown in Figure 1. Applying the resulting gimbal joint command changes the camera viewpoint and produces the next RGB observation, thereby closing the visual feedback loop.

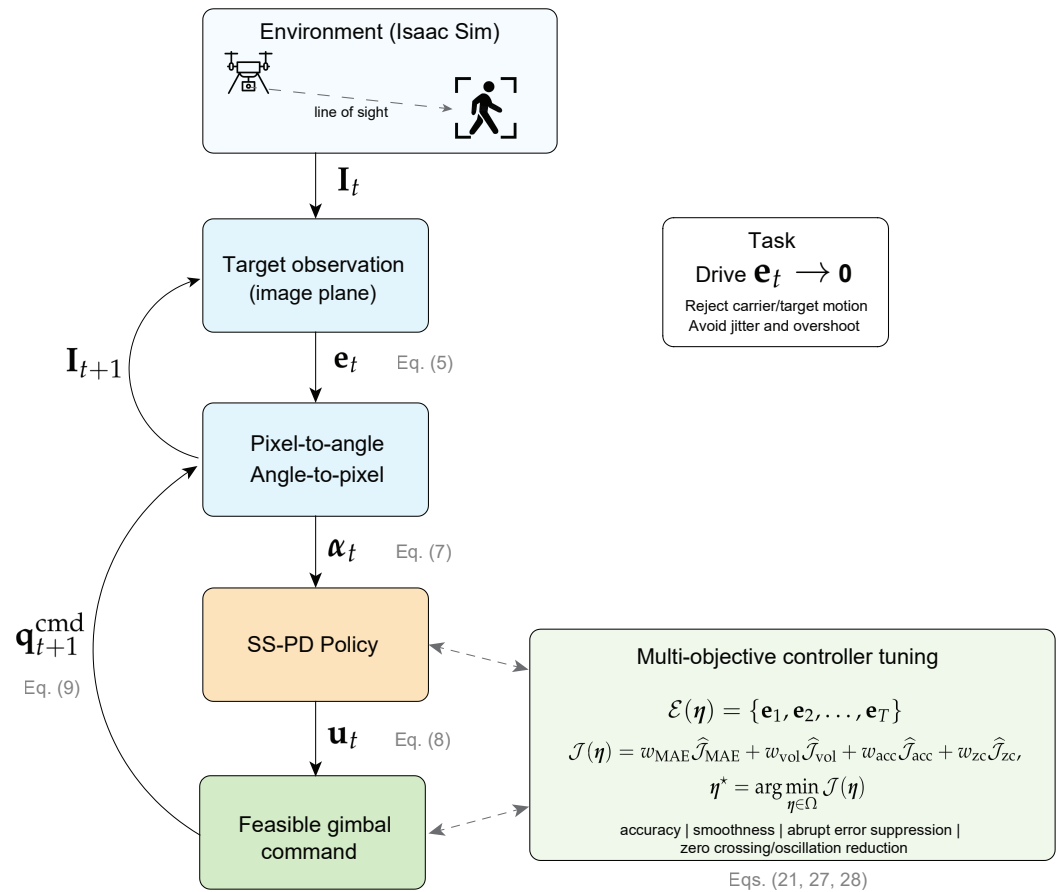


Figure 1. Overview of the proposed vision-based UAV-gimbal tracking framework. The image-plane tracking error is converted into angular feedback and processed by the state-space PD controller to generate feasible gimbal commands. The resulting closed-loop response is used for multi-objective controller tuning.

3.1. Gimbal and Camera Model

The simulated platform consists of a UAV body equipped with a three-axis gimbal and a rigidly mounted RGB camera, as shown in Figure 2. Only the yaw and pitch axes are actively controlled for visual tracking. The controlled gimbal configuration is therefore represented by

$$\mathbf{q}_t = \begin{bmatrix} q_{\psi,t} \\ q_{\theta,t} \end{bmatrix}, \quad (1)$$

where $q_{\psi,t}$ and $q_{\theta,t}$ denote the yaw and pitch angles, respectively. The roll axis is excluded from the target-centering objective and retained according to the simulator configuration.

The camera is modelled using a pinhole perspective projection [18]. The RGB image acquired at time step t is denoted by $I_t \in \mathbb{R}^{H \times W \times 3}$, where W and H denote the image width and height, respectively, and the third dimension represents the three colour channels.

3.2. Target Detection and Temporal Association

The target-observation node uses YOLOv8 to detect human targets in each RGB frame [4,24]. Detection is restricted to the *person* class to suppress irrelevant object categories. Detections below a specified confidence threshold are rejected. All bounding-box coordinates used for image-plane error computation are expressed in the coordinate system of the RGB image used by the controller. ByteTrack associates detections across consecutive frames using a tracking-by-detection strategy [8,25,26]. Once a target has been selected,

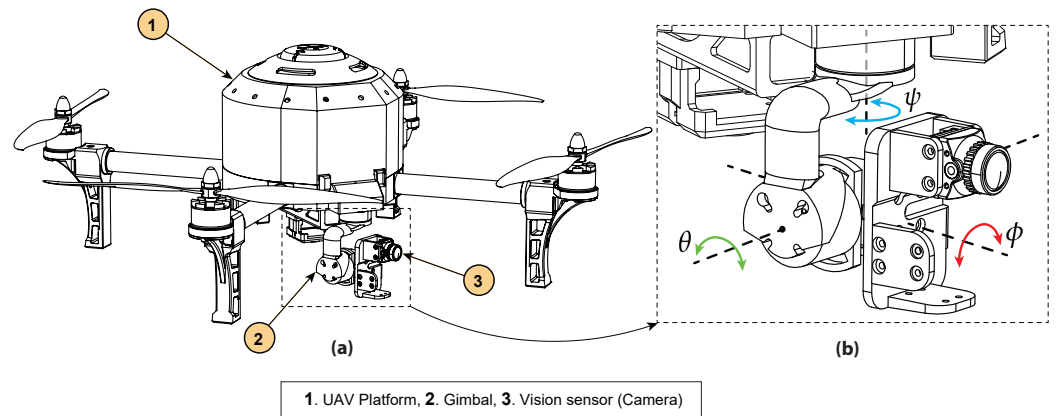


Figure 2. UAV platform equipped with a three-axis gimbal-mounted camera: (a) overall quadrotor platform and (b) close-up of the roll, pitch, and yaw rotational axes. Only the yaw and pitch axes are actively controlled for target centering.

its track identity is stored as the active target ID. The tracking module also retains the last valid image-plane position and a binary validity flag $\nu_t \in \{0, 1\}$, where $\nu_t = 1$ indicates that the selected target has been detected and associated with the active track. If the active target is temporarily absent, the system attempts spatial recovery by selecting the valid detection closest to the last known target position, provided that the displacement lies within a predefined gating radius. If the target cannot be recovered within the specified patience interval, it is declared lost and ν_t is set to zero. This procedure reduces unintended switching between nearby detections and preserves target identity during short occlusions or missed observations.

3.3. Image-Plane Error Extraction

The image-plane target-localization process is illustrated in Figure 3. For a valid observation, the target detector returns the bounding box

$$\mathbf{b}_t = \begin{bmatrix} x_{\min,t} & y_{\min,t} & x_{\max,t} & y_{\max,t} \end{bmatrix}^\top. \quad (2)$$

Its image-plane centre is computed as

$$\mathbf{p}_t = \begin{bmatrix} b_{x,t} \\ b_{y,t} \end{bmatrix} = \begin{bmatrix} \frac{x_{\min,t} + x_{\max,t}}{2} \\ \frac{y_{\min,t} + y_{\max,t}}{2} \end{bmatrix}, \quad (3)$$

and the desired target location is the image centre,

$$\mathbf{p}^* = \begin{bmatrix} c_x \\ c_y \end{bmatrix} = \begin{bmatrix} W/2 \\ H/2 \end{bmatrix}. \quad (4)$$

The corresponding signed image-plane tracking error is defined as

$$\mathbf{e}_t = \begin{bmatrix} e_{x,t} \\ e_{y,t} \end{bmatrix} = \begin{bmatrix} b_{x,t} - c_x \\ c_y - b_{y,t} \end{bmatrix}. \quad (5)$$

The vertical component is inverted because image coordinates increase downwards, whereas the positive pitch direction follows the opposite convention. The magnitude of the target displacement from the image centre is quantified by the radial tracking error $e_{p,t} = \|\mathbf{e}_t\|_2 = \sqrt{e_{x,t}^2 + e_{y,t}^2}$. The signed components determine the required yaw and pitch

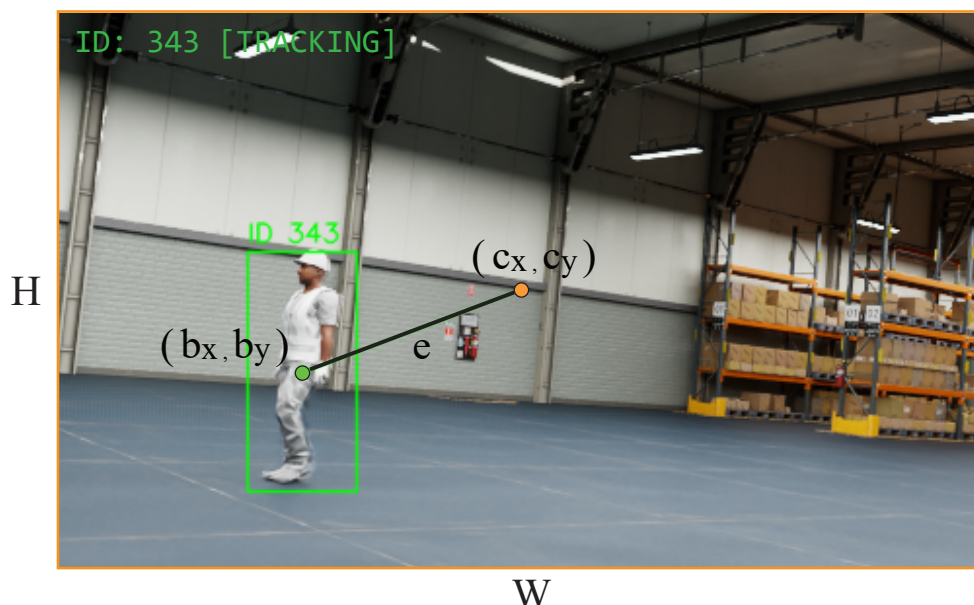


Figure 3. Image-plane target localization. The target detector estimates a bounding box with centre (b_x, b_y) , while (c_x, c_y) denotes the desired image centre. Their signed pixel displacement defines the visual feedback error used for gimbal control.

corrections, while $e_{p,t}$ provides a scalar target-centering measure for controller evaluation and parameter tuning.

3.4. Pixel-to-Angle Conversion

The gimbal controller operates on angular line-of-sight errors rather than directly on pixel displacements, consistent with image-based visual-feedback control [15,17,19]. Let W and H denote the width and height of the RGB image used for image-plane error computation. Let A_x and A_y denote the corresponding horizontal and vertical camera sensor dimensions, and let f denote the focal length. The effective pixel pitches are $s_x = \frac{A_x}{W}$, $s_y = \frac{A_y}{H}$. The signed pixel errors are converted into physical sensor-plane displacements according to $\Delta X_t = e_{x,t}s_x$, $\Delta Y_t = e_{y,t}s_y$. Using the pinhole perspective-projection model [18], the corresponding yaw and pitch line-of-sight errors are

$$e_{\psi,t} = \arctan\left(\frac{\Delta X_t}{f}\right), \quad e_{\theta,t} = \arctan\left(\frac{\Delta Y_t}{f}\right). \quad (6)$$

The angular feedback vector supplied to the controller is therefore

$$\alpha_t = \begin{bmatrix} e_{\psi,t} \\ e_{\theta,t} \end{bmatrix}. \quad (7)$$

This image-based conversion does not require target-depth estimation, three-dimensional reconstruction, or prior knowledge of the target geometry.

3.5. Control Interface and Closed-Loop Operation

The angular feedback vector α_t is processed by the state-space PD controller described in Sec. 4. Its output is the yaw–pitch angular-rate correction

$$\mathbf{u}_t = \Pi_{\eta}(\alpha_t, \bar{\mathbf{x}}_t) = \begin{bmatrix} u_{\psi,t} \\ u_{\theta,t} \end{bmatrix}. \quad (8)$$

Because the Isaac Sim gimbal interface accepts absolute joint-position commands, the angular-rate correction is integrated at each controller sampling instant:

$$\mathbf{q}_{t+1}^{\text{cmd}} = \Pi_{\mathcal{Q}}(\mathbf{q}_t + T_s \mathbf{u}_t), \quad (9)$$

where T_s is the controller sampling period and $\Pi_{\mathcal{Q}}(\cdot)$ denotes projection onto the admissible joint set $\mathcal{Q} = \{\mathbf{q} \mid \mathbf{q}_{\min} \leq \mathbf{q} \leq \mathbf{q}_{\max}\}$. This projection ensures that the commanded yaw and pitch angles remain within the physical limits of the simulated gimbal. Once $\mathbf{q}_{t+1}^{\text{cmd}}$ is applied, the camera orientation is updated and the next RGB frame is acquired. The complete closed-loop information flow is therefore

$$\mathbf{I}_t \rightarrow \mathbf{e}_t \rightarrow \boldsymbol{\alpha}_t \rightarrow \mathbf{u}_t \rightarrow \mathbf{q}_{t+1}^{\text{cmd}} \rightarrow \mathbf{I}_{t+1}. \quad (10)$$

4. Control and Optimization Method

This section presents the internal state-space PD realization used to generate the angular-rate correction $\mathbf{u}[k]$ and the offline NPSO procedure used to determine its parameters. We define t as the image-observation index, while k denotes the controller sampling index. At controller sample k , $\alpha_i[k]$ denotes the most recent angular line-of-sight error available for axis $i \in \{\psi, \theta\}$.

4.1. State-Space PD Realization

Building on the state-space PID formulation of Tan et al. [6], the present work employs a state-space PD realization consisting only of proportional and derivative control actions. The yaw and pitch axes are regulated by two independent controllers with identical structures, as illustrated in Figure 4. The same parameter vector is used for both control channels:

$$\boldsymbol{\eta} = \begin{bmatrix} K_P & K_D & \omega_o \end{bmatrix}^T \in \Omega, \quad (11)$$

where K_P and K_D are the proportional and derivative gains, respectively, ω_o is the observer bandwidth, and Ω denotes the admissible parameter space.

For each controlled axis $i \in \{\psi, \theta\}$, the desired angular line-of-sight error is zero. Accordingly, the reference and measured-output variables are defined as

$$r_i[k] = 0, \quad y_i[k] = -\alpha_i[k], \quad (12)$$

so that the tracking error represented by the block diagram satisfies $r_i[k] - y_i[k] = \alpha_i[k]$.

The two-state observer is defined as

$$\bar{\mathbf{x}}_i[k] = \begin{bmatrix} \bar{x}_{1,i}[k] & \bar{x}_{2,i}[k] \end{bmatrix}^T, \quad (13)$$

where $\bar{x}_{1,i}$ estimates $\dot{y}_i$ and $\bar{x}_{2,i}$ estimates y_i .

The observer feedback coefficients are parameterized by the observer bandwidth as

$$\bar{\beta}_1 = \omega_o^2, \quad \bar{\beta}_2 = 2\omega_o. \quad (14)$$

They define the characteristic polynomial

$$s^2 + \bar{\beta}_2 s + \bar{\beta}_1 = (s + \omega_o)^2, \quad (15)$$

which places both observer poles at $-\omega_o$.

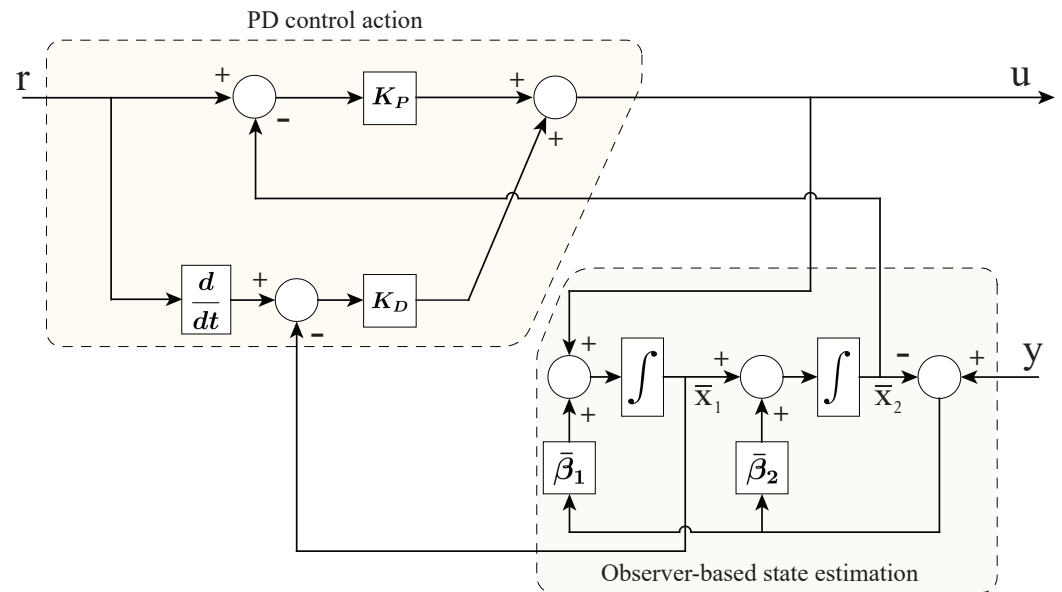


Figure 4. Single-axis state-space PD block diagram; the axis subscript i and controller index k are omitted for clarity. The derivative and proportional actions are evaluated using the observer-based estimates $\bar{x}_1$ and $\bar{x}_2$ respectively. The observer coefficients are parameterized as $\bar{\beta}_1 = \omega_o^2$ and $\bar{\beta}_2 = 2\omega_o$. For the present application, $r_i = 0$ and $y_i = -\alpha_i$.

The continuous observer dynamics, numerically integrated at the controller sampling period, are

$$\dot{\bar{x}}_i = \mathbf{A}_o \bar{\mathbf{x}}_i + \mathbf{B}_o u_i[k-1] + \mathbf{L}_o y_i[k], \quad (16)$$

where

$$\mathbf{A}_o = \begin{bmatrix} 0 & -\bar{\beta}_1 \\ 1 & -\bar{\beta}_2 \end{bmatrix}, \quad \mathbf{B}_o = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{L}_o = \begin{bmatrix} \bar{\beta}_1 \\ \bar{\beta}_2 \end{bmatrix}. \quad (17)$$

Equivalently,

$$\begin{aligned} \dot{\bar{x}}_{1,i} &= u_i[k-1] + \bar{\beta}_1 (y_i[k] - \bar{x}_{2,i}), \\ \dot{\bar{x}}_{2,i} &= \bar{x}_{1,i} + \bar{\beta}_2 (y_i[k] - \bar{x}_{2,i}). \end{aligned} \quad (18)$$

The controller output is then

$$u_i[k] = K_D(\dot{r}_i[k] - \bar{x}_{1,i}[k]) + K_P(r_i[k] - \bar{x}_{2,i}[k]). \quad (19)$$

For $r_i[k] = 0$ and $y_i[k] = -\alpha_i[k]$, this becomes

$$u_i[k] = -K_D \bar{x}_{1,i}[k] - K_P \bar{x}_{2,i}[k]. \quad (20)$$

4.2. Multi-Objective Fitness Formulation

For each candidate parameter vector $\boldsymbol{\eta}$, a complete closed-loop simulation is performed and the resulting image-plane error sequence is recorded as

$$\mathcal{E}(\boldsymbol{\eta}) = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_T\}. \quad (21)$$

Controller tuning involves several competing performance criteria. In this work, these criteria are combined through weighted scalarization into a single objective function optimized by NPSO. Controller performance is evaluated using four complementary measures of target-centering accuracy, temporal variability, abrupt error variation, and oscillation

around the image centre. The mean radial tracking error, here also referred to as the mean absolute error (MAE), is

$$\mathcal{J}_{\text{MAE}} = \frac{1}{T} \sum_{t=1}^T e_{p,t}. \quad (22)$$

It is the mean Euclidean displacement of the target from the image centre. The first difference of the radial error is defined as $\Delta e_{p,t} = e_{p,t} - e_{p,t-1}$, and its standard deviation quantifies error volatility:

$$\mathcal{J}_{\text{vol}} = \sigma(\Delta e_{p,t}). \quad (23)$$

Abrupt variations in the response are characterized using the second difference $\Delta^2 e_{p,t} = e_{p,t} - 2e_{p,t-1} + e_{p,t-2}$, with the corresponding acceleration-like objective

$$\mathcal{J}_{\text{acc}} = \frac{1}{T-2} \sum_{t=3}^T |\Delta^2 e_{p,t}|. \quad (24)$$

This term is derived from the tracking-error sequence rather than directly from the gimbal trajectory and therefore measures abrupt second-order variations in the closed-loop visual response. Oscillation around the optical axis is quantified by

$$\mathcal{J}_{\text{zc}} = N_{\text{zc},x} + N_{\text{zc},y}, \quad (25)$$

where $N_{\text{zc},x}$ and $N_{\text{zc},y}$ denote the zero-crossing counts of the signed errors $e_{x,t}$ and $e_{y,t}$, respectively. Because the four components have different numerical scales, each is normalized before the fitness value is computed. For $m \in \{\text{MAE}, \text{vol}, \text{acc}, \text{zc}\}$, the normalized component is

$$\hat{\mathcal{J}}_m = \min\left(1, \frac{\mathcal{J}_m}{c_m}\right), \quad (26)$$

where the implementation uses $c_{\text{MAE}} = 50 \text{ px}$, $c_{\text{vol}} = 5 \text{ px}$, $c_{\text{acc}} = 2 \text{ px}$, $c_{\text{zc}} = 1000$. The scalar fitness minimized by NPSO is then

$$\mathcal{J}(\boldsymbol{\eta}) = w_{\text{MAE}} \hat{\mathcal{J}}_{\text{MAE}} + w_{\text{vol}} \hat{\mathcal{J}}_{\text{vol}} + w_{\text{acc}} \hat{\mathcal{J}}_{\text{acc}} + w_{\text{zc}} \hat{\mathcal{J}}_{\text{zc}}, \quad (27)$$

and the optimized controller parameters are obtained from

$$\boldsymbol{\eta}^* = \arg \min_{\boldsymbol{\eta} \in \Omega} \mathcal{J}(\boldsymbol{\eta}). \quad (28)$$

The weights in Eq. (27) determine the desired balance between target-centering accuracy, temporal smoothness, abrupt-error suppression, and oscillation reduction. Thus, the proposed formulation uses weighted scalarization to combine the four normalized controller-quality terms into a single PSO fitness [30–32].

4.3. Offline NPSO Search Procedure

Each particle $j \in \{1, \dots, N_p\}$ represents a controller parameter vector $\boldsymbol{\eta}_j^{(k)} = [K_{p,j}^{(k)} \quad K_{D,j}^{(k)} \quad \omega_{o,j}^{(k)}]^\top \in \Omega$, where N_p is the swarm size, $k \in \{0, \dots, N_{\text{it}} - 1\}$ denotes the optimization iteration, $K_{p,j}^{(k)}$ and $K_{D,j}^{(k)}$ are the proportional and derivative gains, respectively, while $\omega_{o,j}^{(k)}$ is the observer bandwidth, and Ω denotes the admissible parameter set. After closed-loop fitness evaluation, the personal best $\mathbf{p}_j^{(k)} \in \Omega$ of particle j and the global best

$\mathbf{g}^{(k)} \in \Omega$ of the swarm are updated. The particle dynamics follow the inertia-weighted PSO formulation in [28], with stability properties discussed in [29]:

$$\tilde{\mathbf{v}}_j^{(k+1)} = w_k \mathbf{v}_j^{(k)} + c_{1,k} \mathbf{r}_{1,j}^{(k)} \odot (\mathbf{p}_j^{(k)} - \boldsymbol{\eta}_j^{(k)}) + c_{2,k} \mathbf{r}_{2,j}^{(k)} \odot (\mathbf{g}^{(k)} - \boldsymbol{\eta}_j^{(k)}), \quad (28)$$

$$\mathbf{v}_j^{(k+1)} = \text{clip}(\tilde{\mathbf{v}}_j^{(k+1)}, -\mathbf{v}_{\max}, \mathbf{v}_{\max}), \quad (29)$$

$$\boldsymbol{\eta}_j^{(k+1)} = \Pi_{\Omega}(\mathbf{g}^{(k)} + \mathbf{v}_j^{(k+1)}), \quad (30)$$

where $\mathbf{v}_j^{(k)}$ is the velocity of particle j , $\tilde{\mathbf{v}}_j^{(k+1)}$ is its unconstrained velocity update, $\mathbf{v}_{\max}$ is the componentwise velocity limit, $\text{clip}(\cdot)$ restricts each velocity component to $[-\mathbf{v}_{\max}, \mathbf{v}_{\max}]$, and $\Pi_{\Omega}(\cdot)$ projects the updated particle onto the admissible set Ω . The coefficients w_k , $c_{1,k}$, and $c_{2,k}$ denote the inertia, cognitive, and social coefficients, respectively, while $\odot$ denotes elementwise multiplication. The entries of the random vectors $\mathbf{r}_{1,j}^{(k)}$ and $\mathbf{r}_{2,j}^{(k)}$ are sampled independently from the uniform distribution $\mathcal{U}(0, 1)$.

Following [7], the global best is retained throughout the search, while the coefficients vary with the normalized iteration index $\rho_k = k/N_{\text{it}}$, where N_{it} is the total number of optimization iterations. The coefficient schedules are

$$w_k = 0.9 - 0.5\rho_k, \quad c_{1,k} = 2(1 - \rho_k^{1.5}), \quad c_{2,k} = 2\rho_k^{1.5}. \quad (31)$$

Particle positions and initial velocities are sampled uniformly within Ω and $\pm\mathbf{v}_{\max}$, respectively. The search uses 20 particles and 11 iterations, yielding 220 closed-loop fitness evaluations. The parameter bounds and velocity limits are

$$\Omega = [7, 14] \times [0, 0.70] \times [20, 1000], \quad \mathbf{v}_{\max} = \begin{bmatrix} 0.7 & 0.07 & 100 \end{bmatrix}^{\top}. \quad (32)$$

The search bounds were anchored to the Ziegler–Nichols baseline and refined through preliminary closed-loop tests. Specifically, $K_P \in [7, 14]$ and $K_D \in [0, 0.70]$ bracket the baseline values ($K_P \approx 7.68$, $K_D \approx 0.48$). The K_P lower bound retains a discernible proportional response, while $K_D = 0$ allows the search to admit purely proportional action. The upper bounds were selected from preliminary tests beyond which persistent oscillation was observed. The observer range $\omega_o \in [20, 1000]$ brackets the ZN-SSPD value $\omega_o \approx 632$ rad/s, obtained from $\omega_o = 2N/T_d$ with $N = 20$, following [6]. The lower bound was selected to avoid excessively slow state estimation, whereas the upper bound is half the explicit-Euler stability limit, $2/T_s = 2000$ rad/s, for $T_s = 1$ ms. The search terminates after the fixed iteration budget without a convergence-based stopping criterion or fixed random seed. The best parameter vector is then fixed for online control.

4.4. Online Control Execution and Data Logging

The controller is executed with a sampling period of $T_s = 0.001$ s. Because perception and control may operate at different rates, the most recent valid angular feedback measurement is retained using a sample-and-hold mechanism until a new target observation becomes available. When $v_t = 1$, the current angular error is supplied to the yaw and pitch controllers. If $v_t = 0$, the visual tracking update is suspended and the gimbal initially holds its last valid command. If the target remains unavailable beyond the specified recovery interval, the gimbal returns to its default stabilization configuration. For each closed-loop simulation, the recorded signals include the signed horizontal and vertical pixel errors, radial tracking error, angular line-of-sight errors, controller outputs, commanded yaw and pitch positions, observer states, and target-validity flag. These signals are used to

evaluate the fitness components, compare the controller variants, and analyse the resulting accuracy–smoothness trade-offs.

5. Experimental Setup

The proposed framework is evaluated in NVIDIA Isaac Sim using two complementary experiments. The first examines continuous target tracking under simultaneous carrier and target motion, whereas the second isolates the transient response to a large initial image-plane offset. All controller variants are evaluated using the same perception–control architecture, sampling configuration, target-observation procedure, and gimbal constraints described in the preceding sections.

5.1. Execution Environment

The experiments are conducted on a desktop workstation running NVIDIA Isaac Sim and ROS 2. The principal hardware and software specifications are listed in Table 1.

Table 1. Hardware and software environment used for the simulation experiments.

Component	Specification
Operating system	Linux Mint 22.1
Kernel	Linux 6.8.0-62-generic
CPU	AMD Ryzen 7 9700X
Memory	31 GiB RAM
GPU	NVIDIA GeForce RTX 5070 Ti
GPU driver	570.133.07
ROS 2 distribution	Jazzy
Python	3.12.3
Simulator	NVIDIA Isaac Sim 5.0.0

5.2. Evaluation Scenarios

Table 2 summarizes the two evaluation scenarios. Scenario 1 represents continuous visual tracking under time-varying line-of-sight disturbances. The UAV carrier follows a prescribed trajectory while the animated human target walks through the scene. This experiment assesses the ability of the controller to maintain target centering while rejecting disturbances caused by both carrier and target motion.

Scenario 2 evaluates the transient response independently of continuous external motion. The carrier and target remain stationary, while the initial camera orientation places the target approximately 200 pixels horizontally and 150 pixels vertically from the image centre. This configuration is used to examine convergence, overshoot, and residual oscillation following a large initial tracking error.

Table 2. Evaluation scenarios used for controller assessment.

Item	Scenario 1	Scenario 2
Evaluation objective	Continuous tracking	Transient response
UAV carrier	Prescribed sinusoidal motion	Static
Target	Walking human	Static
Initial image-plane error	Determined by relative motion	Approximately (200, 150) px
Evaluation duration	100 s	5 s
Primary assessment	Accuracy and response quality	Overshoot and settling behaviour

To ensure a consistent comparison, all evaluated controller configurations are initialized from the same scenario-dependent state and subjected to identical carrier and target trajectories. The optimized controller parameters and fitness weights used in Experiments 2–7 are reported in Table 3. Experiment 2 uses MAE alone, while Experiments 3–5

augment MAE with one additional penalty. Experiment 6 assigns equal weight to all four normalized objectives. Experiment 7 evaluates a reweighted combined objective with $w_{\text{MAE}} = w_{\text{zc}} = 1$, $w_{\text{vol}} = w_{\text{acc}} = 0.5$, thereby reducing the relative emphasis on temporal variation while retaining full weighting on target centering and zero-crossing suppression. The evaluated configurations use state-space PD controllers with K_P , K_D and the observer bandwidth ω_o selected by NPSO.

Table 3. NPSO-tuned controller parameters and fitness weights for Experiments 2–7.

Exp.	Fitness	K_P	K_D	ω_o	w_{MAE}	w_{vol}	w_{acc}	w_{zc}
2	MAE	13.2213	0.2255	487.1265	1	0	0	0
3	MAE + Volatility	12.3629	0.1225	932.0433	1	1	0	0
4	MAE + Zero-crossings	12.8675	0.1359	956.2599	1	0	0	1
5	MAE + Acceleration	12.0321	0.1477	901.5414	1	0	1	0
6	Combined (equal weights)	9.6348	0.1508	600.2994	1	1	1	1
7	Combined (reweighted)	10.2877	0.1202	818.4926	1	0.5	0.5	1

5.3. Carrier Motion Profile

In Scenario 1, the UAV is treated as a kinematic carrier whose prescribed motion generates deterministic disturbances at the gimbal mounting frame. Its world-frame position is defined by

$$x(t) = 10 \sin(31.416t) - 12, \quad (33)$$

$$y(t) = -18.6, \quad (34)$$

$$z(t) = 3.6, \quad (35)$$

while its orientation is varied about the world-frame y -axis according to

$$\varphi_x(t) = 0, \quad (36)$$

$$\varphi_y(t) = 30 \sin(31.416(t + 4)), \quad (37)$$

$$\varphi_z(t) = 0. \quad (38)$$

These trajectories are applied through the Isaac Sim Action Graph and remain fixed across all Scenario 1 experiments. Consequently, every controller is evaluated under the same repeatable line-of-sight excitation. No carrier or target motion is imposed in Scenario 2. Instead, the initial gimbal orientation is selected to generate the prescribed image-plane offset, after which the controller is allowed to recenter the target.

5.4. Evaluation Protocol

For each controller configuration, the complete closed-loop simulation is executed over the duration specified in Table 2. The diagnostic signals recorded during each run follow the logging procedure described in Sec. 4.4.

Scenario 1 is evaluated using the mean radial tracking error, error volatility, acceleration-like error variation, and horizontal and vertical zero-crossing counts defined in Eqs. (22)–(25). The normalized values defined in Eq. (26) are reported for both controller comparison and internally during NPSO tuning. For Scenario 2, transient performance is assessed from the signed horizontal and vertical image-plane errors. The principal quantities of interest are the peak excursion beyond the image centre, the subsequent decay of the tracking error, and the presence of residual oscillations.

For each component $i \in \{x, y\}$, let $e_{i,0}$ denote the initial signed error. Its overshoot is defined as the maximum excursion beyond the image centre in the direction opposite to the initial error:

$$O_i = \max_{1 \leq t \leq T} \{\max(0, -\text{sgn}(e_{i,0})e_{i,t})\}, \quad i \in \{x, y\}. \quad (39)$$

The maximum image-plane overshoot and corresponding percentage overshoot are

$$i^* = \arg \max_{i \in \{x, y\}} O_i, \quad O_{\max} = O_{i^*}, \quad O_{\%} = 100 \frac{O_{\max}}{|e_{i^*,0}|}. \quad (40)$$

This definition treats an exact zero-valued sample as an image-centre crossing and assigns zero overshoot to a component that does not cross the image centre.

For Experiment k , the percentage change relative to the MAE-only baseline in Experiment 2 is

$$\Delta O_k = 100 \frac{O_{\max,k} - O_{\max,2}}{O_{\max,2}}, \quad (41)$$

where $\Delta O_k < 0$ indicates an overshoot reduction. Together, the two scenarios provide complementary assessments of continuous disturbance rejection and large-error transient recovery.

6. Experimental Results

This section reports the performance of the proposed vision-based gimbal tracking framework in the two evaluation scenarios defined in Sec. 5. Experiment 1 compares the classical Ziegler–Nichols PD controller with its state-space counterpart. Experiments 2–7 subsequently examine how different NPSO fitness terms influence tracking accuracy, temporal error variation, oscillatory behaviour, and large-offset transient recovery.

6.1. Experiment 1: ZN-PD versus ZN-SSPD

The ZN-PD and ZN-SSPD controllers use the same baseline proportional and derivative gains obtained using the classical Ziegler–Nichols tuning method [27]. The comparison therefore isolates the effect of the state-space realization before NPSO-based parameter tuning is introduced. Both baseline controllers use $K_P = 7.68$ and $K_D = 0.48$.

6.1.1. Scenario 1: Dynamic Tracking

Figure 5 compares the normalized Scenario 1 metrics for the ZN-PD and ZN-SSPD controllers. Relative to ZN-PD, the ZN-SSPD controller reduces the normalized MAE from 0.240 to 0.225, the acceleration-like error metric from 0.216 to 0.208, and the zero-crossing metric from 0.217 to 0.183. However, the normalized volatility increases slightly from 0.067 to 0.072. Thus, under the tested Ziegler–Nichols gains, the state-space realization improves three of the four evaluated metrics, with a modest increase in sample-to-sample error volatility.

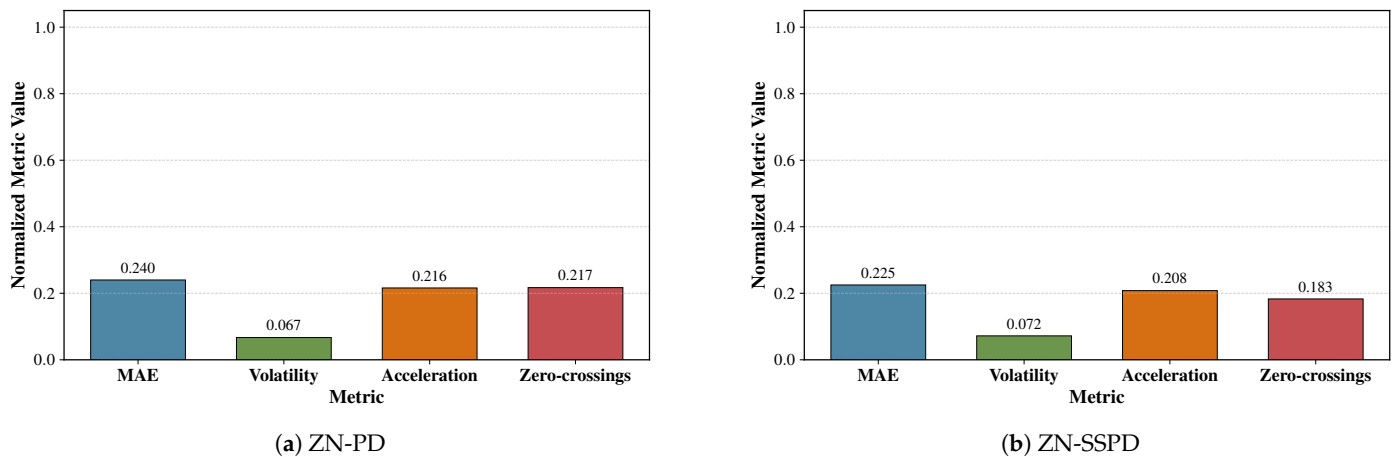


Figure 5. Normalized Scenario 1 performance metrics for the ZN-PD and ZN-SSPD controllers. The evaluated quantities are the mean radial tracking error, reported as MAE, error volatility, acceleration-like second-order variation of the radial error, and the combined horizontal and vertical zero-crossing count.

6.1.2. Scenario 2: Step Response

The large-offset transient responses are compared in Figure 6. The maximum overshoot decreases from 36.37 px for ZN-PD to 24.27 px for ZN-SSPD, corresponding to a reduction of approximately 33.3%. The associated percentage overshoot decreases from 23.39% to 16.28%. For the tested baseline gains, the state-space realization therefore produces a smaller image-centre crossing after the large initial tracking error. This baseline comparison motivates the subsequent investigation of optimization-based parameter selection.

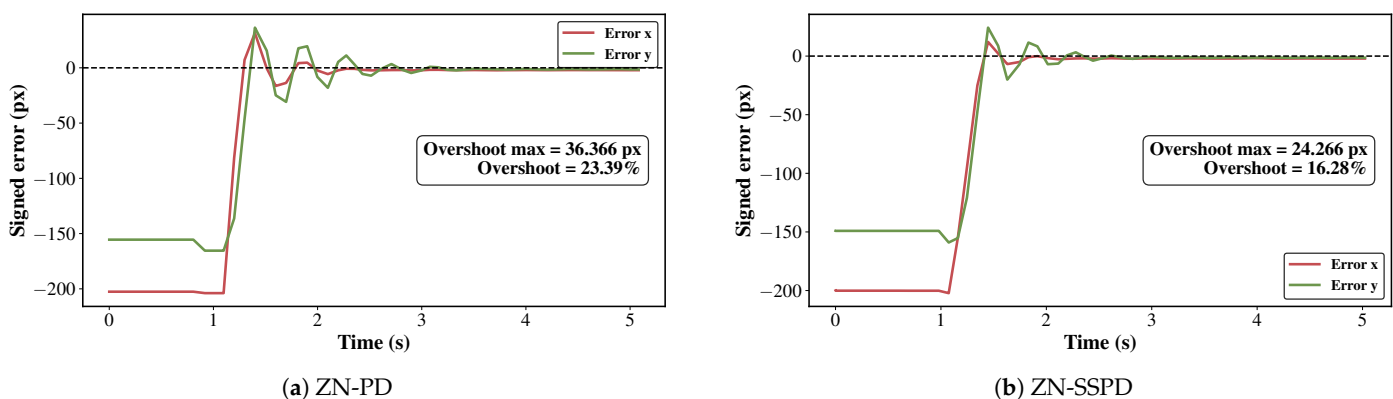


Figure 6. Transient recovery from a large initial image-plane offset using the ZN-PD and ZN-SSPD controllers. The signed horizontal and vertical tracking errors are shown for both controllers. The ZN-SSPD realization reduces the maximum overshoot from 36.37 px (23.39%) to 24.27 px (16.28%).

6.2. Experiments 2–7: Influence of NPSO Fitness Terms and Weighting

Experiments 2–7 evaluate the influence of the NPSO fitness formulation on the state-space PD controller. Experiment 2 uses MAE-only optimization and serves as the NPSO baseline. Experiments 3–5 augment MAE with volatility, zero-crossing, and acceleration-like penalties, respectively. Experiment 6 combines all four terms with equal weights, whereas Experiment 7 reweights the combined objective to reduce the relative contribution of the volatility and acceleration terms.

6.2.1. Scenario 1: Normalized Tracking Metrics

Figure 7 compares the normalized Scenario 1 metrics for the six NPSO-tuned configurations. Experiment 2 achieves the lowest MAE (0.162) but also the largest acceleration-like (0.247) and zero-crossing (0.400) metrics. Adding volatility, zero-crossing, or acceleration penalties in Experiments 3–5 reduces temporal variation and oscillation at the cost of higher MAE. Among these, Experiment 5 yields the lowest acceleration-like metric (0.120).

The equal-weight combined objective in Experiment 6 produces the lowest acceleration-like metric (0.011) and shares the lowest volatility (0.005), but has the largest MAE (0.224) and a zero-crossing value of 0.267. Reweighting the combined objective in Experiment 7 reduces MAE to 0.190 and zero crossings to 0.150, the lowest among the NPSO configurations, while increasing volatility to 0.069 and acceleration-like variation to 0.097. Thus, no formulation dominates across all metrics; the resulting trade-off depends on both the selected objective terms and their relative weights.

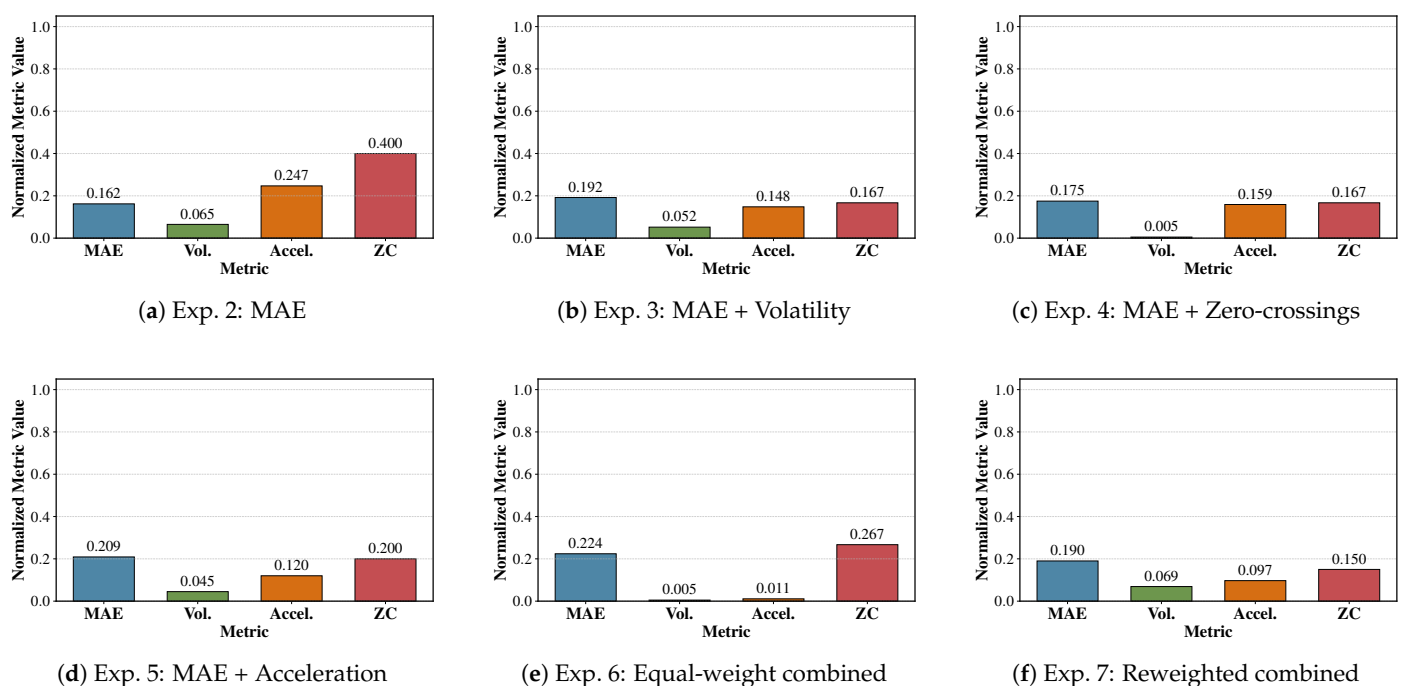


Figure 7. Normalized Scenario 1 performance metrics for the NPSO fitness formulations considered in Experiments 2–7. The evaluated quantities are the mean radial tracking error, reported as MAE, error volatility, acceleration-like second-order variation of the radial error, and the combined zero-crossing count. The results show that the different fitness formulations produce distinct trade-offs among target-centering accuracy, temporal error variation, and image-centre crossings.

6.2.2. Scenario 2: Overshoot Response

Figure 8 and Table 4 compare the large-offset transient responses of Experiments 2–7. The maximum image-plane overshoot and corresponding percentage overshoot are computed according to Eqs. (39) and (40), while the percentage change relative to the MAE-only baseline is obtained from Eq. (41).

Relative to Experiment 2, all augmented formulations reduce the maximum overshoot. Experiments 3–7 achieve 29.12, 33.31, 27.03, 29.30, and 31.10 px, respectively, compared with 37.78 px for Experiment 2. Experiment 5 provides the largest reduction (28.5%), while the equal-weight and reweighted combined objectives reduce overshoot by 22.4% and 17.7%, respectively. Thus, although both combined formulations improve the transient response relative to the MAE-only baseline, neither matches the overshoot reduction achieved by the acceleration-augmented objective.

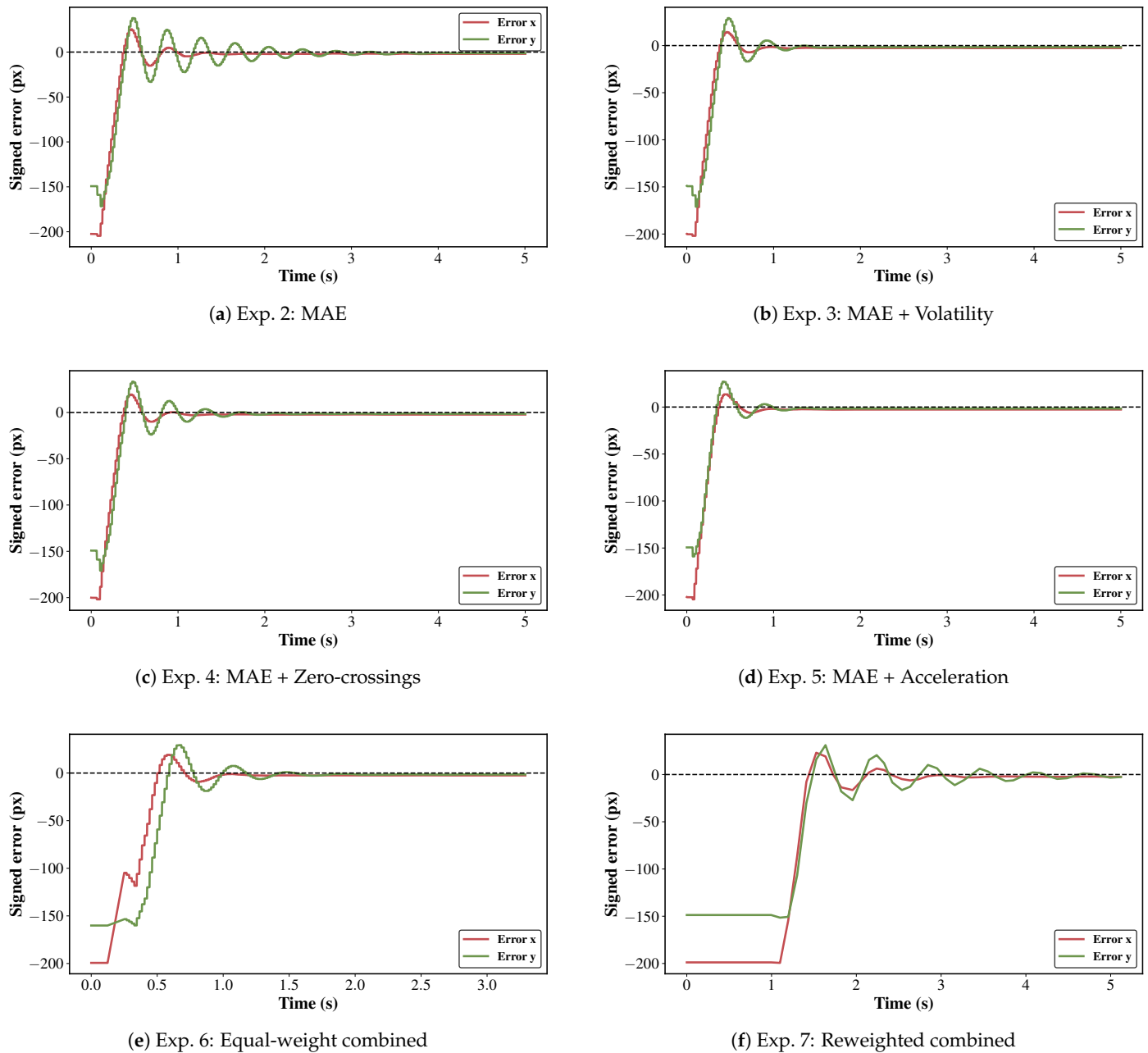


Figure 8. Transient recovery from a large initial image-plane offset for the NPSO fitness formulations evaluated in Scenario 2. The signed horizontal and vertical tracking errors are shown for Experiments 2–7, illustrating the influence of the different objective terms on overshoot, residual oscillation, and convergence.

Table 4. Scenario 2 overshoot results for Experiments 2–7. The percentage change is computed relative to the MAE-only baseline in Experiment 2. Lower overshoot values are better.

Exp.	Fitness function	Overshoot [px]	Overshoot [%]	Change vs. Exp. 2
2	MAE	37.78	25.31	–
3	MAE + Vol.	29.12	19.55	–22.9%
4	MAE + ZC	33.31	22.32	–11.8%
5	MAE + Acc.	27.03	18.09	–28.5%
6	Combined (equal weights)	29.30	18.30	–22.4%
7	Combined (reweighted)	31.10	20.91	–17.7%

7. Discussion

Taken together, the experiments show that a single average-error measure does not fully characterize vision-based gimbal tracking performance. Target-centering accuracy remains the primary control objective, but the temporal evolution of the visual error also affects overshoot, oscillation around the image centre, and the perceived stability of the camera view. The results therefore indicate that controller quality should be assessed using complementary steady-state and transient measures.

7.1. Effect of the State-Space PD Structure

The comparison between ZN-PD and ZN-SSPD shows that the state-space realization can improve the closed-loop response under the tested baseline gains. As shown in Figure 5, ZN-SSPD reduces the normalized MAE, acceleration, and zero-crossing metrics compared with ZN-PD, while producing a slightly higher volatility value. In the step-response case, Figure 6 shows that ZN-SSPD also reduces the maximum overshoot.

These results indicate that the observer-based formulation improves error regulation and transient crossing behaviour for the tested configuration. This observation is consistent with image-based airborne-gimbal studies showing that the controller structure and disturbance-rejection mechanism strongly influence line-of-sight tracking performance [20]. However, the improvement is not uniform across all metrics. The slight increase in volatility suggests that the state-space structure alone is insufficient to guarantee an optimally smooth response. The controller still requires systematic gain and observer-bandwidth tuning, especially when tracking accuracy and smoothness must be balanced simultaneously.

Alternative gimbal-control approaches address these requirements using model predictive control [23], self-tuning fuzzy PID control [33], adaptive feedback linearization [34], and filter-assisted stabilization [35]. In contrast, the proposed approach retains the comparatively simple PD structure while using an observer-based realization and offline multi-criteria tuning. The present results therefore do not establish superiority over these alternative methods, but show that the state-space realization improves the tested classical PD baseline and provides a suitable structure for subsequent NPSO-based parameter optimization.

7.2. Accuracy–Smoothness Trade-Off

The NPSO experiments reveal a clear trade-off between target-centering accuracy and visual-error smoothness. The MAE-only objective in Experiment 2 achieves the lowest average radial error in Scenario 1, but also produces the largest acceleration-like variation and zero-crossing count, as shown in Fig. 7. This behavior reflects the competing nature of the objectives, where improving one criterion may degrade another [30], and is consistent with the sensitivity of PSO-tuned controllers to the selected fitness function [31].

Adding volatility, acceleration-like, and zero-crossing penalties generally reduces temporal error variation or image-centre crossings at the cost of increased MAE. The comparison between Experiments 6 and 7 further shows that this trade-off depends on the relative objective weights. Reweighting the combined objective improves MAE and yields the lowest zero-crossing metric, but increases volatility, acceleration-like variation, and transient overshoot relative to the equal-weight formulation. Thus, objective weighting provides a direct mechanism for shifting the closed-loop response according to application priorities.

For observation tasks prioritizing visual stability, increased centering error may be acceptable in exchange for smoother error evolution, whereas precision-tracking tasks may favor stronger MAE weighting. Similar application-dependent adjustment of gimbal-controller behavior has also motivated self-tuning approaches [33].

7.3. Effect of Acceleration and Zero-Crossing Penalties

The acceleration-like and zero-crossing objectives capture different properties of the visual response. The former is derived from the second difference of the radial image error and therefore measures abrupt second-order changes in image-plane displacement rather than physical gimbal acceleration or mechanical jerk. Its strong overshoot performance in Experiment 5 suggests that suppressing such variations also improves recovery from large initial errors.

The zero-crossing term penalizes sign changes in the horizontal and vertical image errors and is therefore more directly associated with oscillation about the optical axis. It reduces image-centre crossings, although its sensitivity to small noise-induced sign changes may motivate a deadband or hysteresis criterion for hardware implementation.

The equal-weight combined objective in Experiment 6 achieves the lowest acceleration-like variation and shares the lowest volatility, whereas the reweighted formulation in Experiment 7 yields the lowest zero-crossing metric. The associated increase in volatility, acceleration-like variation, and overshoot confirms that objective weighting shifts the closed-loop trade-off rather than producing a uniformly superior controller.

7.4. Implications for UAV Gimbal Tracking

The appropriate fitness formulation depends on the operating requirements of the UAV–gimbal system. A stronger MAE weighting is suitable when maintaining the target close to the optical axis is the dominant objective. A zero-crossing penalty is useful when repeated image-centre crossings are undesirable, while an acceleration-like error penalty is advantageous when large-offset overshoot and abrupt visual-error changes must be reduced. The full multi-objective formulation should therefore be viewed as a flexible weighting framework rather than as a universally optimal configuration. Its weights can be adapted to different mission modes, including precision tracking, stable observation, or rapid target reacquisition. Maintaining line-of-sight and image stability is also a central requirement in inertially stabilized optical platforms [21]. Moreover, because the present smoothness measures are derived from the image-error sequence, smoother error evolution does not necessarily imply lower physical gimbal acceleration, control effort, or mechanical loading. Gimbal-control studies based on model predictive control have similarly considered the system response under dynamic disturbances [23], while filter-assisted stabilization methods highlight the importance of suppressing undesirable motion components [35]. Physical gimbal motion, control effort, and mechanical loading should therefore be evaluated directly when the controller is implemented on hardware.

7.5. Limitations and Sim-to-Real Considerations

The present evaluation is limited to controlled simulation. The UAV carrier is modelled kinematically, and the imposed motion profiles act as deterministic line-of-sight disturbances rather than full aerodynamic or force-based perturbations. Visual-domain differences between simulated and physical environments may affect perception performance [36], while mismatches between simulated and physical dynamics may alter the transferred controller behaviour [37]. These sim-to-real limitations are particularly relevant to vision-based UAV systems [38]. Real-world effects such as camera calibration error, motion-dependent rolling-shutter distortion [39], actuator backlash, communication delay, vibration, wind disturbance, and full UAV attitude dynamics are not explicitly modelled.

The evaluation also considers one prescribed dynamic trajectory and one large-offset initial condition. Consequently, the reported comparisons do not yet establish performance across a broad distribution of carrier motions, target speeds, viewing geometries, occlusions, or initial tracking errors. The reported smoothness measures are derived from the

image-plane tracking error rather than directly from the gimbal trajectory. Hardware evaluation should therefore include angular velocity, angular acceleration, control effort, joint saturation, image jitter, detector confidence, and target-ID stability. Future work should also examine axis-specific controller gains, because the present shared parameter vector may not remain optimal when the yaw and pitch dynamics differ. Validation on a physical UAV–gimbal platform is required to establish hardware-level robustness. Such experiments should include randomized carrier and target trajectories, varying initial offsets, detector latency, temporary target loss, mechanical backlash, and synchronization between the camera, perception, and control nodes. Adaptive objective weighting may also be investigated for missions in which the preferred balance among centering accuracy, visual smoothness, oscillation suppression, and transient recovery changes during operation.

8. Conclusion

This paper presented a vision-based UAV gimbal tracking framework combining YOLOv8–ByteTrack target observation, image-plane visual feedback, state-space PD control, and NPSO-based gain and observer-bandwidth tuning. The framework was evaluated in NVIDIA Isaac Sim under dynamic tracking and step-response scenarios.

The state-space PD realization improved the tested ZN-PD baseline by reducing average tracking error, zero crossings, acceleration-like error variation, and overshoot, with a slight increase in volatility. The NPSO experiments showed that the fitness formulation strongly influences the closed-loop response. MAE-only tuning provided the best target centering, whereas additional volatility, acceleration, and zero-crossing penalties shifted the response toward smoother or less oscillatory behavior. The MAE + acceleration objective achieved the lowest transient overshoot, while reweighting the combined objective yielded the lowest zero-crossing metric, confirming that both objective selection and relative weighting govern the resulting trade-off.

Overall, average image-plane error alone is insufficient to characterize UAV gimbal tracking performance. Multi-criteria tuning provides a practical mechanism for balancing centering accuracy, visual-error smoothness, oscillation suppression, and transient response. Future work will consider randomized trajectories, varying target speeds, hardware validation, and adaptive objective weighting for mission-dependent operation.

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Abbreviations

The following abbreviations are used in this manuscript:

UAV	Unmanned Aerial Vehicle
RGB	Red–Green–Blue
ROS 2	Robot Operating System 2
PID	Proportional–Integral–Derivative
PD	Proportional–Derivative
SS-PID	State-Space Proportional–Integral–Derivative
SS-PD	State-Space Proportional–Derivative
ZN	Ziegler–Nichols
ZN-PD	Ziegler–Nichols Proportional–Derivative
ZN-SSPD	Ziegler–Nichols State-Space Proportional–Derivative
PSO	Particle Swarm Optimization
NPSO	Novel Particle Swarm Optimization
MAE	Mean Absolute Error
ZC	Zero Crossing
DoF	Degree of Freedom
URDF	Unified Robot Description Format
CPU	Central Processing Unit
GPU	Graphics Processing Unit
RAM	Random Access Memory

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